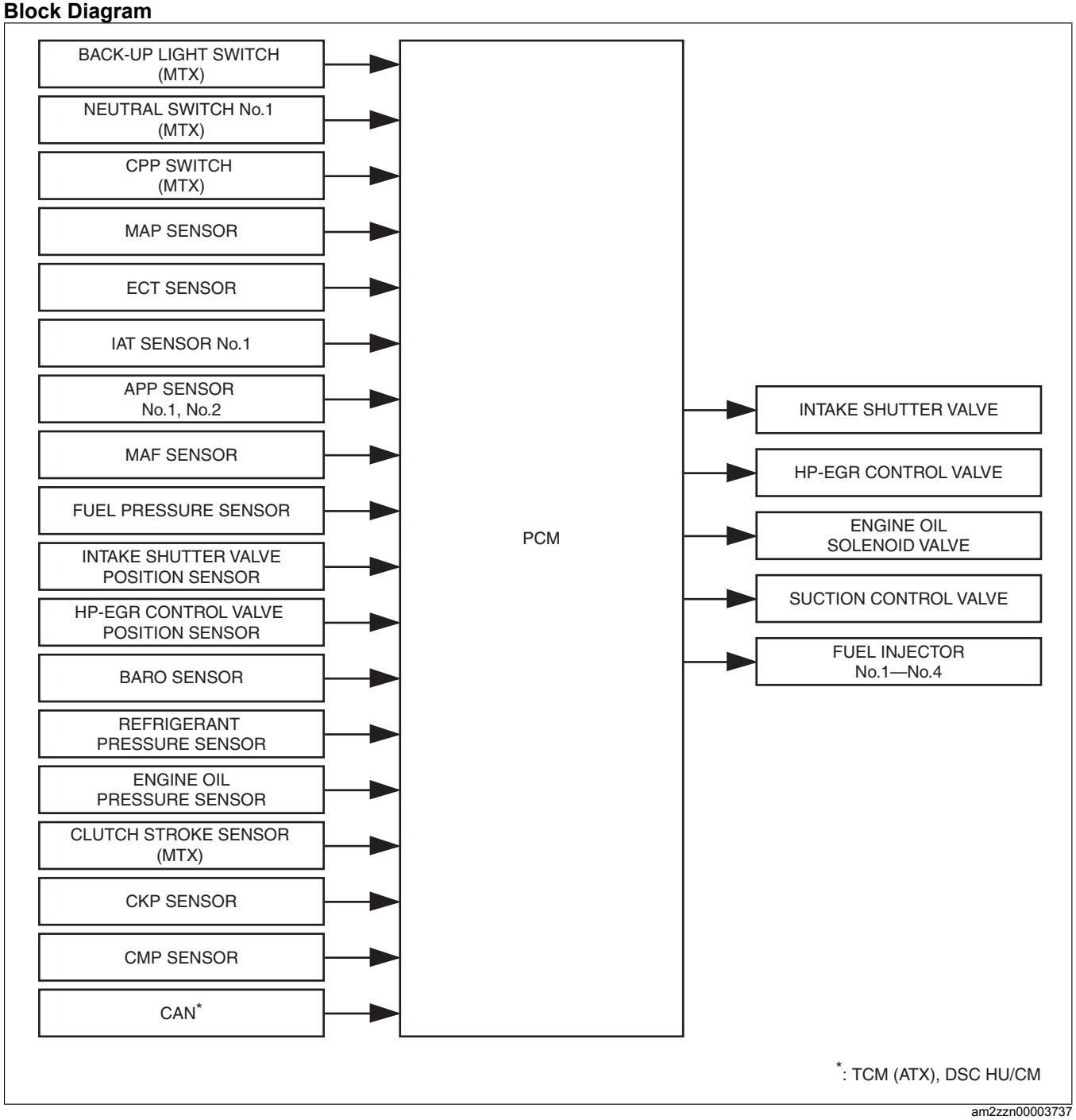
## IDLE SPEED CONTROL [SKYACTIV-D 1.5]

id0140z5952300

### Outline

- The PCM controls the fuel injection amount so that the target idle speed according to the engine operation conditions is obtained.

### Block Diagram




am2zzn00003737

### Operation

- The fuel injection amount is controlled so that the actual idle speed is close to the target idle speed.
- The target idle speed is set under various conditions and consists mainly of the following control items.

| Item | Control description | Target idle speed |
|---|---|---|
| Engine speed control during normal operation | The actual idle speed conforms to the target idle speed during normal idling. | **775 rpm** (ATX)/**800 rpm** (MTX) |
| Engine speed control during warm-up | Approaches the target engine speed during warm-up based on the input signal from ECT sensor and IAT sensor. | **775—1,500 rpm** |

| Item | Control description | Target idle speed |
|---|---|---|
| Engine speed control during fuel injection amount learning | During fuel injection amount learning, the engine speed is controlled so that it matches the target engine speed. | **During auto-learning control**<br>**800 rpm**<br>**During manual learning control**<br>**800—1,500 rpm** |
| DPF regeneration control | Controls the engine speed during DPF regeneration control so that it matches the target engine speed. | **Auto DPF regeneration control**<br>**800—925 rpm**<br>**Compulsory DPF regeneration control**<br>**1,750 rpm** * |

* : If there is incorrect combustion of soot during diesel particulate filter regeneration, the exhaust gas temperature increases which may cause damage to the diesel particulate filter. In this case, post injection is stopped and the engine speed increases to **2,500 rpm**. Due to this, low temperature exhaust gas is sent to the oxidation catalytic converter at high speed and the diesel particulate filter is cooled to prevent damage.

- The A/C operation information from the A/C pressure sensor is input to prevent the idle speed from decreasing by A/C load. The fuel injection amount is corrected to maintain idle speed when the A/C load is applied.
  In addition, the fuel injection amount is corrected to prevent a decrease in speed when the electrical load is applied.